Gravitation

Curie Circle Research Group

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The purpose of this handout is to gain a practical understanding of gravitation through Kepler's Laws both conceptually and manipulatively. Enjoy!

§1 Introduction

You probably know what a force is. In a rough sense, an object attracts another. Simple enough. These "forces" follow a set of rules that were deemed Newton's Laws through Newton's own observation and were shown to be correct in most cases, as for the exceptions, you'll see them in a higher-level physics class.

Definition 1.1. Newton's Laws are summarized as follows:

- Newton's First Law: An object in rest stays at rest, and an object in motion stays in motion with the same speed and in the same direction unless acted upon by an unbalanced external force.
- Newton's Second Law: The acceleration of an object is directly proportional to the net force acting on it and inversely proportional to its mass.
- Newton's Third Law: For every action, there is an equal and opposite reaction. Forces always occur in pairs.

This is all well and good. An apple falls onto Newton's head with an acceleration to the force acting on it (gravity) and hits Newton (quite hard) and Newton feels a reaction force (with a bump on his head). It is natural then to wonder whether these laws extend into higher dimensions of size.

For massive objects, do these laws still hold? We may contend these are true for a large rock, a house, and an Eiffel tower. But, say, does it hold for a planet? For such a huge object, does it follow a certain path in motion or can it stop? What happens with a collision or around another object?

These questions start with the formulation of a gravitational force. A force that is defined to be the law of gravity between two objects. This was formulated by Newton as follows.

Definition 1.2. The gravitational force between two objects m_1 and m_2 is equal to

$$F = \frac{Gm_1m_2}{r^2}$$

where r is the distance between m_1 and m_2.

At first glance, this formula is calculational, but there are other interesting parts. For instance, what exactly is G?

Let us first understand it in terms of units. Rearranging to get G, we have:

$$G = \frac{Fr^2}{m_1 m_2}$$

The units of force $[F]$ is Newtons N or $\frac{kg \cdot m}{s^2}$. Hence, the units of G are

$$[G] = \frac{[F] \cdot [r]^2}{[m_1] \cdot [m_2]} = \frac{kg \cdot m}{s^2} \cdot \frac{m^2}{kg^2} = \frac{m^3}{kg \cdot s^2}$$

This G is known as the gravitational constant: $G = 6.67 \cdot 10^{-11} Nm^2 kg^2$ to be precise. Essentially, this the scale of the gravitational attraction between two objects.

Now imagine that we exist as an object with a large shell. A shell can be thought of as a cover that has mass on the outside edge. Since we exist without another mass, we may think there is no gravitational force. This is both right and wrong. These theorems attributed to Newton can shed some light on this matter.

Theorem 1.3

The gravitational acceleration within a hollow, uniformly dense spherical shell is 0 for $r < R$ where R represents the radius of the shell. However, the gravitational acceleration outside the shell is as if all the mass of the shell were concentrated at its center for $r \geq R$.

A pictorial description is below:

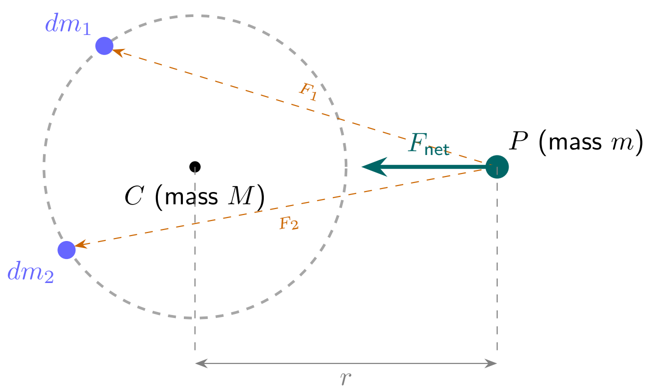


Figure 1: Case 1: P is **outside** the shell ($r > R$). The shell exerts a net force $F = GMm/r^2$, exactly as if all mass M were concentrated at the centre C. Individual forces F_1, F_2 from opposite patches do not cancel — their resultant always points toward C.

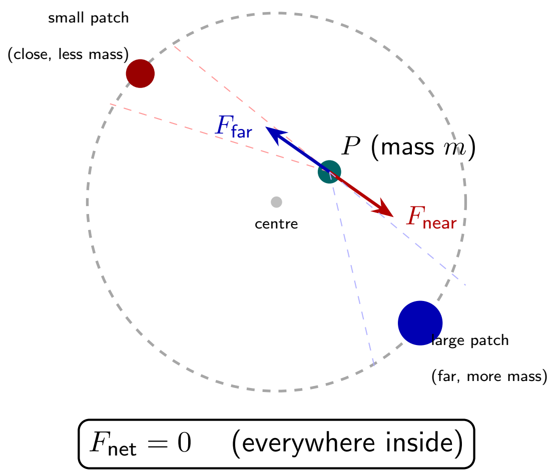


Figure 2: Case 2: P is **inside** the shell ($r < R$). Although P is off-centre, the larger solid angle toward the far side intercepts proportionally more mass, and the two effects cancel exactly. The net gravitational force from the shell is **zero** at every interior point.

While the formal derivation of these theorems requires some difficult techniques, we can intuitively understand this theorem.

When $r > R$, effectively a mass m is outside of the sphere. Thus if we consider the forces from the masses outside, they don't cancel out. However, the vertical components do cancel because they are opposite to each other. Since they don't cancel, we can reason that it should be nonzero. To prove the exact relationship lies a bit more effort, but we can think of the situation as adding a bunch of opposite vectors (two points whose chord is perpendicular to the line connecting point P to C) whose resultant (the addition of the vectors) lies on $\overline{PC}$.

For the case of $r < R$, the point mass P may be closer to one side than the other. If it is in the "perfect" center of the circle, then clearly the force vectors cancel out, so the net force is 0. Hence, in the case that it is closer to one side than the other, the side closer provides a larger force (since the mass corresponding to the arc is larger) than that of the farther side. However, the side farther from P has more distance from P than the side closer to P. These effects of strength over length adequately balance each other out to get 0.

§2 Kepler's Laws

While this formulation of the force of gravity between objects does answer some questions that are mentioned, we still don't have an answer for the path of these planetary objects. Through Newton's Laws, Johannes Kepler explained what exactly happens to these masses.

Lemma 2.1

Kepler's Laws for a general orbit are

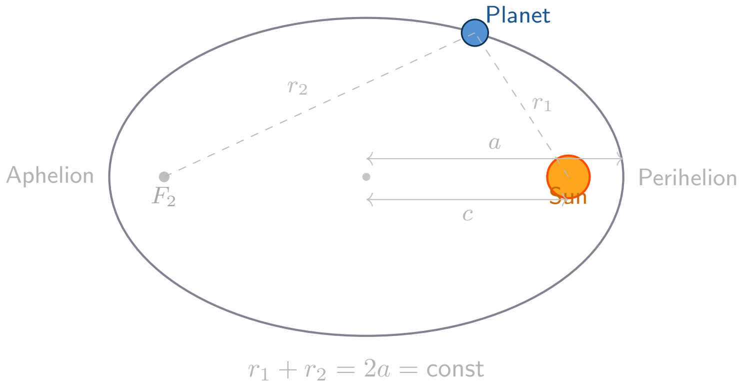
- The trajectories of planets are sections, with a focus at the Sun. Bound orbits are ellipses, circles if the eccentricity is 1.
- The Trajectories sweep out equal areas in equal times.
- When the orbit is bound, the period T and semimajor axis a obey $T^2 \propto a^3$.

A pictorial view of these Laws is below:

Kepler's Three Laws of Planetary Motion

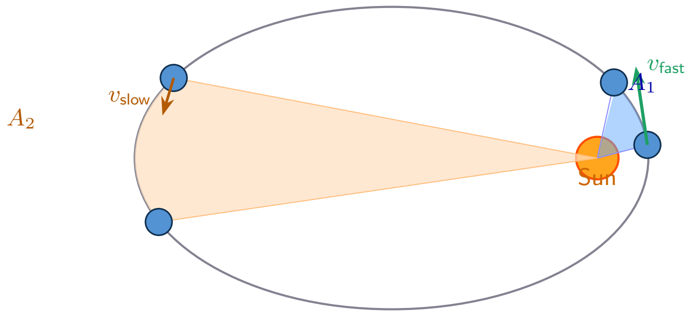
Law I

Each planet orbits the Sun in an ellipse with the Sun at one focus.



Law II

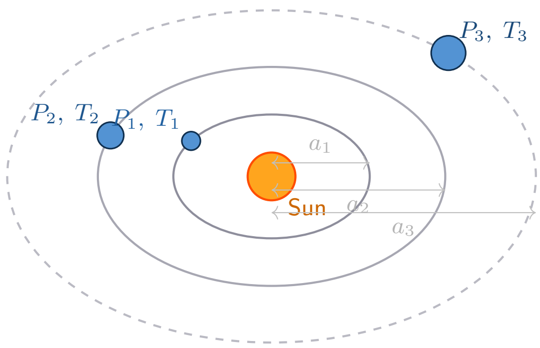
A line from the Sun to a planet sweeps equal areas in equal times.



$$\frac{L}{2m} = \text{const}$$

Law III

The square of a planet's orbital period is proportional to the cube of its semi-major axis.

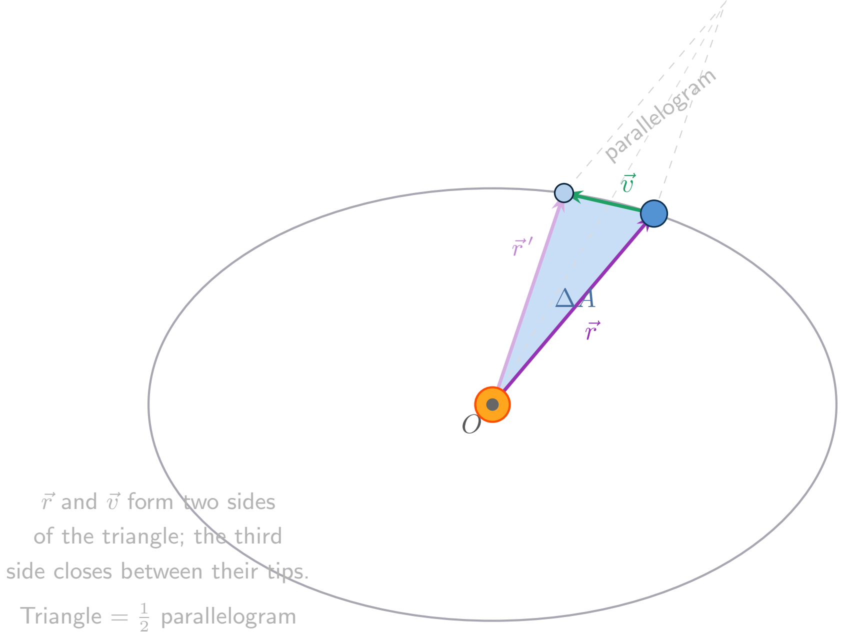


$$\frac{T^2}{a^3} = \text{const} \quad \Longleftrightarrow \quad T^2 \propto a^3$$

While these Laws are complex, we can get an intuitive understanding for each. The full derivation is complex, and you will see them in a higher level physics class. However, we can still get an appreciation of them using simple techniques.

Kepler's 1st Law may be unmotivated for those of you who have not worked with ellipses before, so take a leap of faith that it holds.

Kepler's 2nd Law can be explained intuitively like so: take a part of the area that the mass goes over a period of time. This $\frac{A}{t}$ is equal to $\frac{1}{2}r \cdot v$ because this is the only dimensionally correct interpretation (or think of it as a "right" triangle) and the factor of $\frac{1}{2}$ is there as if it were not, it would be a parallelogram, which is not necessarily correct.



$$\boxed{\Delta A = \tfrac{1}{2} \cdot |\vec{r}| \cdot |\vec{v}| \cdot \sin\theta \;=\; \frac{L}{2m} = \text{const}}$$

The ΔA corresponds to a change in time. We can "assume" that $\sin\theta$ tends to 1, so this is effectively saying that the change in area is equal to $\frac{1}{2}rv$. Recall that the angular momentum is $L = m \cdot v \cdot r$, so $v \cdot r = \frac{L}{m}$, which implies

$$\Delta A = \frac{L}{2m}$$

but the angular momentum is constant throughout an object's orbit, so ΔA is constant. Do note that this is an intuitive argument, and there are some details to be ironed out to make it rigorous. But, this will work for us.

Finally, let's consider Kepler's Third Law in that $T^2 \propto a^3$. We can imagine a circle (as sort of a cheat code), with $a = R$. Imagine a point mass is in orbit R (directly above the planet). Let the velocity of the object in orbit be v. Then, the time to complete an orbit is

$$T = \frac{2\pi r}{v}$$

We can also directly find v through the formula connecting the centripetal force to the gravitational force:

$$m\frac{v^2}{r} = \frac{GMm}{r^2} \implies v^2 = \frac{GM}{r}$$

Hence

$$T^2 = \frac{4\pi^2 r^2}{v^2} = 4\pi^2 r^2 \cdot \frac{r}{GM} \propto r^3$$

§3 Problems

Take a partner and pick the 2 to 3 problems in this list that look the most interesting to do.

Problem 3.1. Alex is on a planet covered with loose rock on its surface. Alex is interested in creating a space station in a geostationary orbit around this planet. Is it possible to Alex to do so, regardless of the planet's rotation speed?

Problem 3.2. Alex has rephrased Kepler's Laws as follows

- (a) the orbits of planets are elliptical with one focus at the sun.
- (b) a line connecting the sun and a planet sweeps out equal areas in equal times.
- (c) the square of the period of a planet's orbit is proportional to the cube of its semimajor axis.

Alex knows that these laws hold when the force of gravity is proportional to $\frac{1}{r^2}$. However, he wonders whether any of these would hold if the force of the gravity is proportional to $\frac{1}{r^3}$. Can you help Alex?

Problem 3.3. Alex is in a spaceship orbiting a planet. Alex is prone to falling asleep on the wheel since there are no space cops to catch him. At those times, the ship must fire its thrusters to avoid falling into the atmosphere. Where should the thrusters point (the direction where exhaust is fired) to give the greatest increase in the maximum height of its orbit?

Problem 3.4. The supermassive black hole at the center of the galaxy Axodus is called Gigantamax $A*$ with a mass of 8×10^{36} kilograms. It takes 200 million years for the solar system to orbit once around the galaxy.

Under the assumption that the only object in this galaxy is Gigantamaz $A*$, find the distance of the galazy from the center.

Previous explorers found the value to be $2 \cdot 10^{20}$ meters. Assuming this value is correct and given the approximation we made, does the value reached in the previous part make sense?

Problem 3.5. Adapted from F=MA A spherical mist of miniscule particles in space has a uniform density p_0 and a radius r_0. The gravitational acceleration of free fall at the surface of the mist due to the mass of the mist is g_0. A process occurs that causes the mist to suddenly grow to a radius $2R_0$, while maintaining a uniform density, which may or may not be equal to the original density. What is the new gravitational acceleration of free fall at a point r_0 away from the center of the mist?

Don't be scared by the process, look at gravity.

Problem 3.6. The Δv in spaceflight is defined as the magnitude of the change in velocity needed during a particular maneuver in space. What is the Δv needed to escape Planet X's gravitational pull from low-planet orbit? Let Planet X have mass m and radius r.

Problem 3.7. Imagine that at a given point in time, the Plant X, Plant Y, and Planet Z lie on the same line (collinear). How many days will it be before they are all collinear again? Let Planet X have period 100 days, planet Y have period 200 days, and Planet Z have period 400 days.

Problem 3.8. Imagine a drill is dug down to the center of the a Planet X. A mass m is placed a depth d into the drill at a distance $r = r_x - d$ from the center of the Earth where r_x is the radius of planet X. What is the weight of the object at this point? Solve in terms of the mass of Planet X, radius of Planet X, mass of object, and distance of the mass into the drill.

Problem 3.9. On Planet Robo of mass M and radius R, robots have dug out a spherical area of radius $\frac{R}{2}$ as shown. Originally, Planet Robo had mass M. Find the acceleration due to gravity at the top of the cavern and the acceleration at the bottom.

§4 Reduced Mass

Having gone through the previous problems, you may ask the following question: "Given several planets, how exactly can we configure a coordinate system that handles each planet simplistically?" Obviously we can plot a dot as the origin, but then considering each planet from that origin seems messy.

This question will be answered in the following discussion. But let us first consider the following example.

Example 4.1

Consider two planets, both of mass m, say A and B. Alex holds A in place (yes he's that strong), and B performs a circular orbit of radius r and period t. Then, Alex lets go of both planets, and they perform circular orbits simultaneously of radius $\frac{r}{2}$ about their center of mass. What is the new period of this motion?

Let us consider each situation alone. In the former, planet A acts as the origin of our system, and planet B merely orbits planet A. Let Planet B have velocity v, then we can write the following through a centripetal acceleration calculation:

$$\frac{mv^2}{r} = \frac{Gm^2}{r^2} \implies v = \sqrt{\frac{Gm}{r}}$$

In the latter situation, the only thing that changes is that the center of mass differs, resulting in a new radius of $\frac{r}{2}$. The planets still interact solely through gravitational forces, which is separated by a distance of $2 \cdot \frac{r}{2}$ or r.Therefore:

$$\frac{mv^2}{\frac{r}{2}} = \frac{Gm^2}{r^2} \implies v = \frac{1}{\sqrt{2}}\sqrt{\frac{Gm}{r}}$$

In this case the velocity is smaller by a factor of $\frac{1}{\sqrt{2}}$. The motion is also a factor of 2 smaller due to the radius. Hence the period is $\frac{t}{\sqrt{2}}$.

This is all well and good, but we may wonder if there exists a setup that makes things easier. The first situation was fine because planet A was the origin, so it essentially wasn't there. The second one was also not too much trouble, but could there be a better way?

The answer is yes. The following fact may seem unintuitive, but have trust that it holds.

Lemma 4.2

Consider two objects of mass m_1 and m_2 with positions r_1 and r_2. Their relative position is $r_2 - r_1$ under the appropriate gravitational forces of the system. For the purposes of computationally computing characteristics of this system, velocities, forces, radii, etc, we may replace this system with a single mass μ in the same system. μ satisfies the equation

$$\frac{1}{\mu} = \frac{1}{m_1} + \frac{1}{m_2}$$

This new system with the reduced mass μ is isomorphic to the other coordinate system.

The word "isomorphic" may seem unwieldy, but it just states that nothing changes from the previous system to this one. All the forces, distances, etc when calculated properly work out to be the same.

Now, let us see how to apply this lemma in the same example.

We can consider the relative position $r_a - r_b$ between planets A and B. Notice that these planets orbit in a circle of radius r. Using the lemma, we can define μ to be

$$\mu = \frac{1}{\frac{1}{m} + \frac{1}{m}} = \frac{m}{2}$$

which acts in a circular orbit of radius r as the problem describes.

As our "reduced mass" system is the "same" as the original system, the forces between masses stays the same, so the centripetal force is $\frac{Gm^2}{r^2}$. Therefore, the reduced mass μ fulfills the equation:

$$\frac{\mu v^2}{r} = \frac{Gm^2}{r^2} \implies v = \sqrt{\frac{2Gm}{r}}$$

Notice that it is r and not $\frac{r}{2}$ because this reduced mass has an orbit of a circle of radius r. Since the speed is bigger by a factor of $\sqrt{2}$, but the length of the orbit is the same, the period becomes $\frac{t}{\sqrt{2}}$.

Problem 4.3. This is a hard problem, but very fun.

Alex has two objects that act under a net force that is attractive. No torques are generated, so effectively these two objects interact through a central force. If one of the objects is held still (again Alex is strong) and the other is released, the period of collision is T. If instead Alex holds the other object, the period of collision is T'. If instead both objects were released from rest, how long does is the period of collision?